Deep Research of Copilot - Round 2 Analysis

Aggregate Response State and Deep Third-Cumulant Hypotheses

Bottom Line

Derived: Any successful replacement of old sources must preserve a realized odd field of approximately O(n^2) degrees of freedom. The cost comes from retaining the realized odd structure, not merely rank information. Aggregate old sources before transport rather than after transport.

Q1: Aggregate Response Fit

Represent old content as T = sum_a sym3(G_a tensor v_a). Fit directly against the contraction/query family used by the chain rather than tensor entries. Use randomized query operators and alternating least squares. Estimated useful range k ≈ 128-256, but viability requires aggressive collapse after source aggregation. A practical target is k <= 32.

Cost Model

Per component transport: G <- BGB^T with approximately 2 units per layer. If merged content can be represented with k=16 components, total deep-tier cost becomes roughly 32 units per layer and may approach leaderboard territory.

Primary Falsifier

At one layer (e.g. L10), merge all sources with age >= 5, fit increasing k, and measure relative error epsilon_U. If achieving 2 percent error requires k > 32, reject the approach.

Q2A: Propagated Covariance of Odd Fields

Carry one realized odd field X in R^(n x n) together with a closed two-point covariance model. Update X using X'=(W■W)XW^T. Cost roughly 2 units per layer.

Q2B: Birth-Index-First Contraction

Aggregate birth-column statistics before neuron contractions using M=sum w2_j p_j p_j^T and N=sum w2_j a_j p_j^T. Propagate both as BMB^T and BNB^T. Test reconstruction error of D21 from {M,N}.

Q2C: Response-Function Formulation

Propagate response vertices R^(m) and reconstruct cumulant effects through diagrammatic expansions. Cost O(n^2) state and roughly 2-10 units per layer. Low prior probability.

Q3: Most Likely Leader Strategy

Leaders probably do not carry old sources individually. Instead they may maintain a small number of aggregated O(n^2) odd fields, fitted once at the age gate and transported recursively thereafter.

Strong Negative Argument

If aggregate fitting requires k >= 64, then little credible room remains. Current measurements imply the signal is realized, odd, not recoverable from even statistics, not low-rank in Hadamard space,

not closed on slices, and needed across all births and late layers.